

DISC-NET ML Workshop

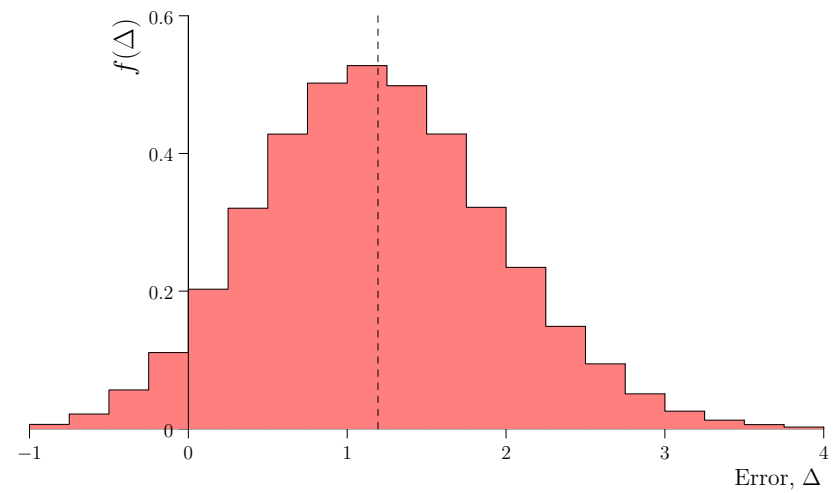
Lessons from Machine Learning Mistakes



Data sets, reproducibility and metrics

Outline

1. **Data Sets**
2. Reproducibility
3. Metrics



History of Doing Things Badly

- We learn from our mistakes
- The history of machine learning is full of mistakes
- That is, doing thing poorly leading to misunderstanding and confusion in the literature

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- That is, doing thing poorly leading to misunderstanding and confusion in the literature
- This still happens, all the time, but we have learned to do things better

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- In deep learning these have to be large
- Of course, to be useful to commoners like you and me, we also want them to be easy to handle

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MNIST



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- It had 60 000 training images (roughly 6000 per digit)
- 10 000 test images
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Sins of MNIST

- MNIST isn't that hard
- The digits are already centred and rescaled so it is not that interesting
- Many practitioner's would run test multiple times and state their best performance
- For most good networks, MNIST is so easy that the errors are where the scaling and centring went wrong or the digits are unintelligible
- Ignoring this practitioners would start over-fitting on the test set

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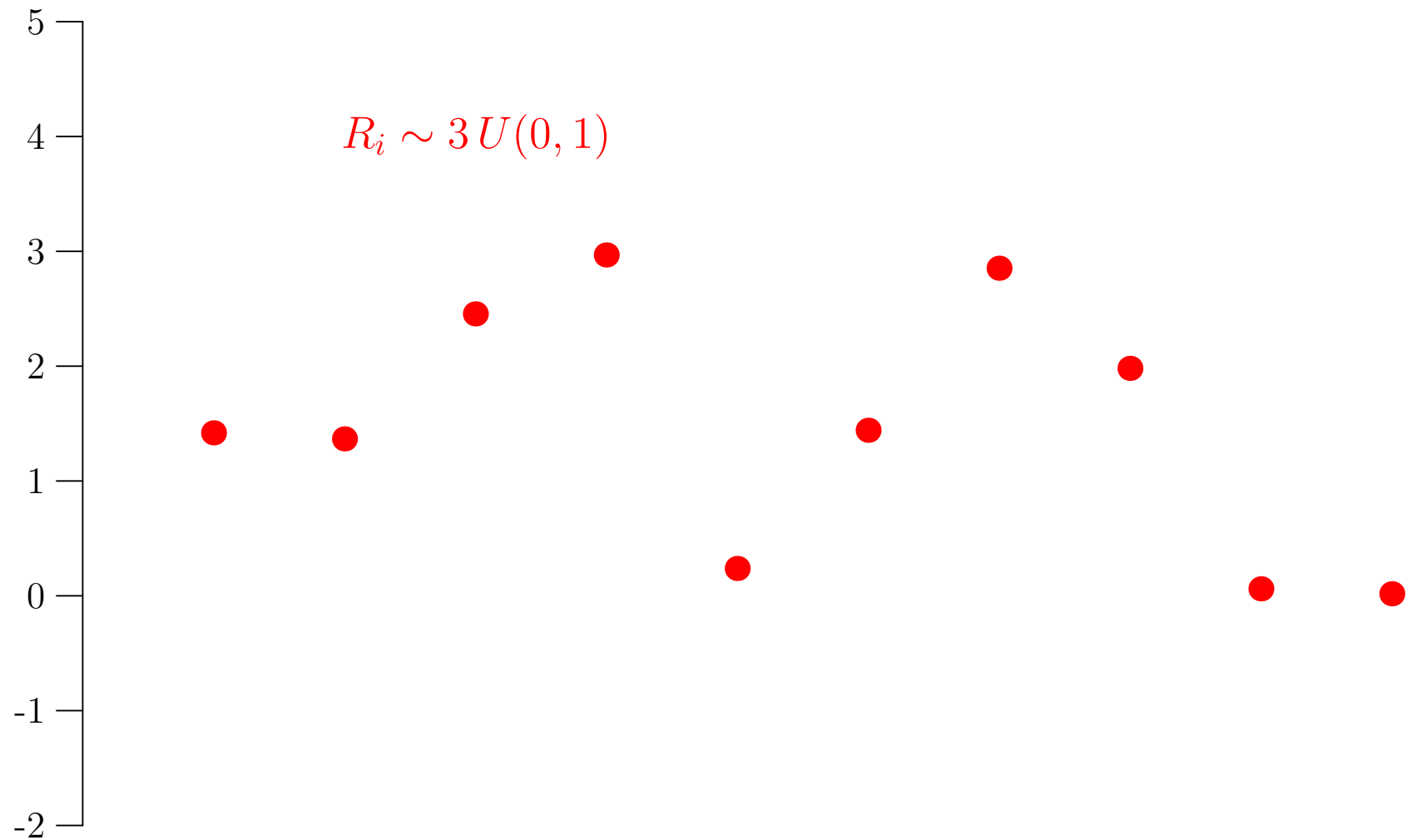
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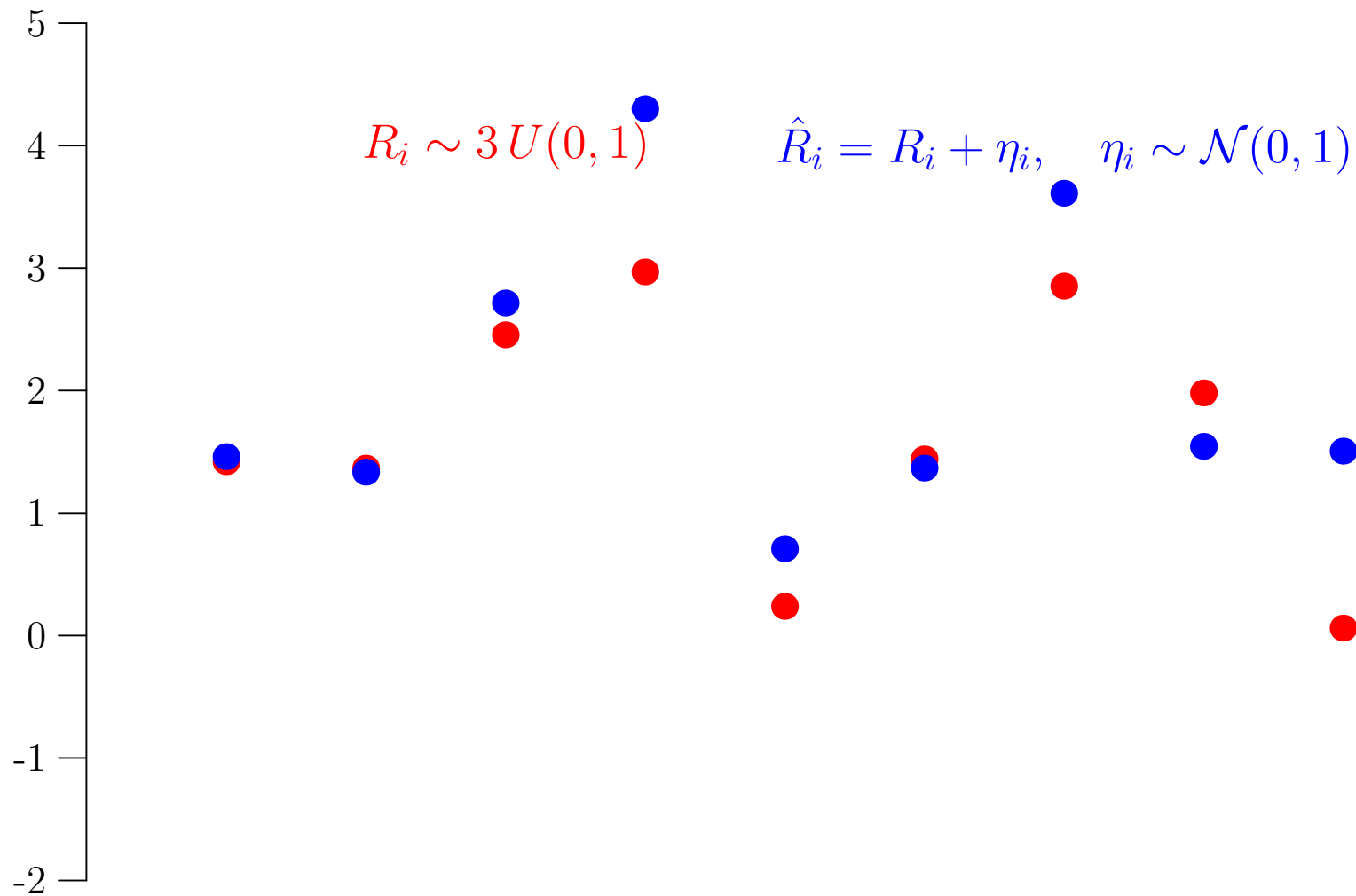
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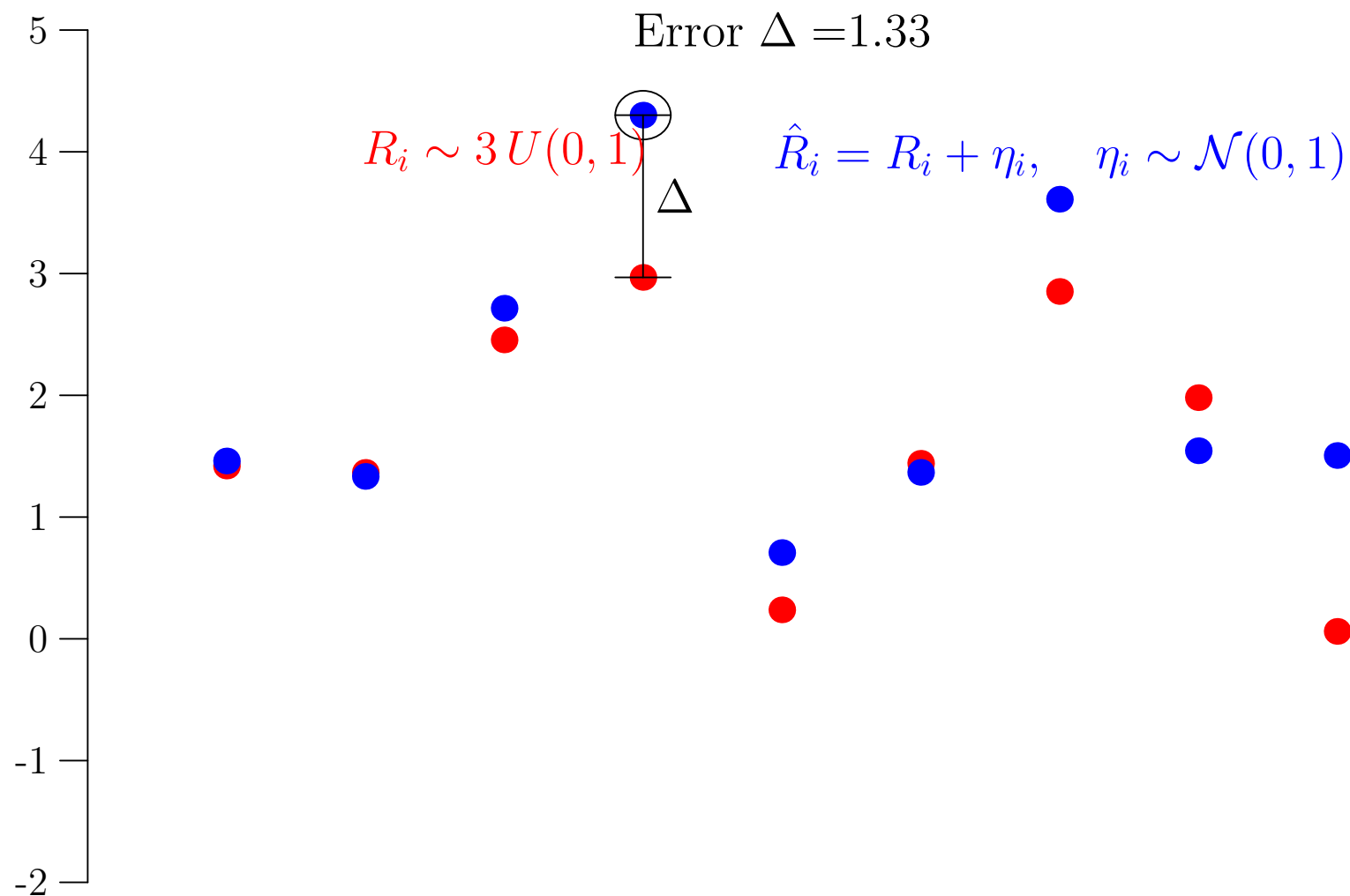
The Over-fitting Game



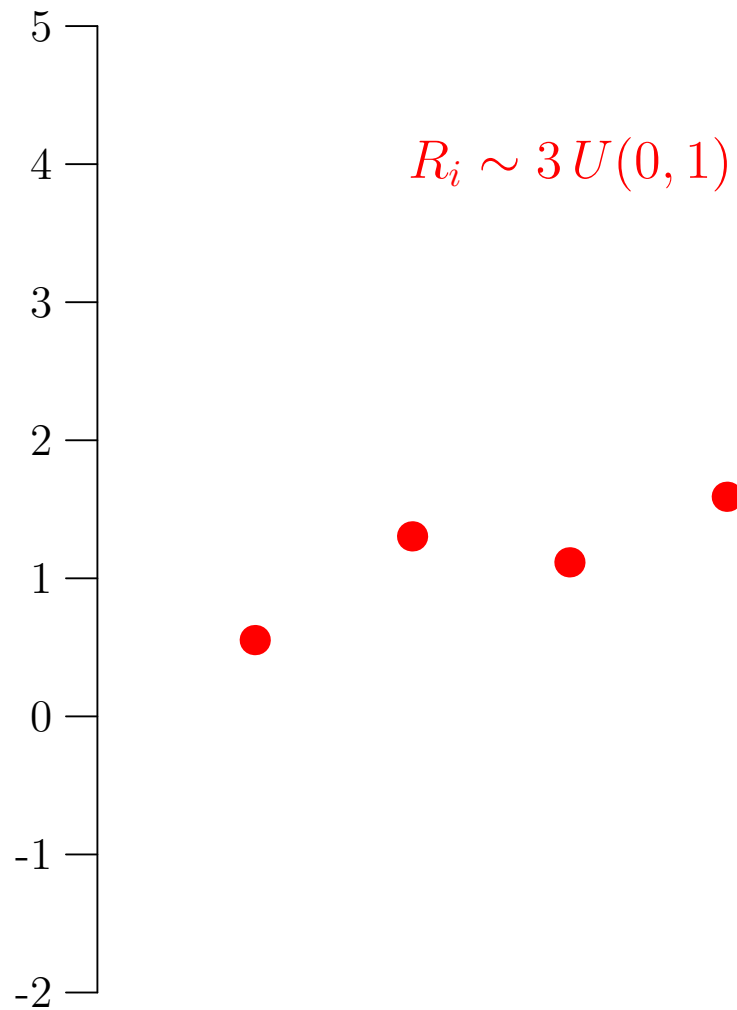
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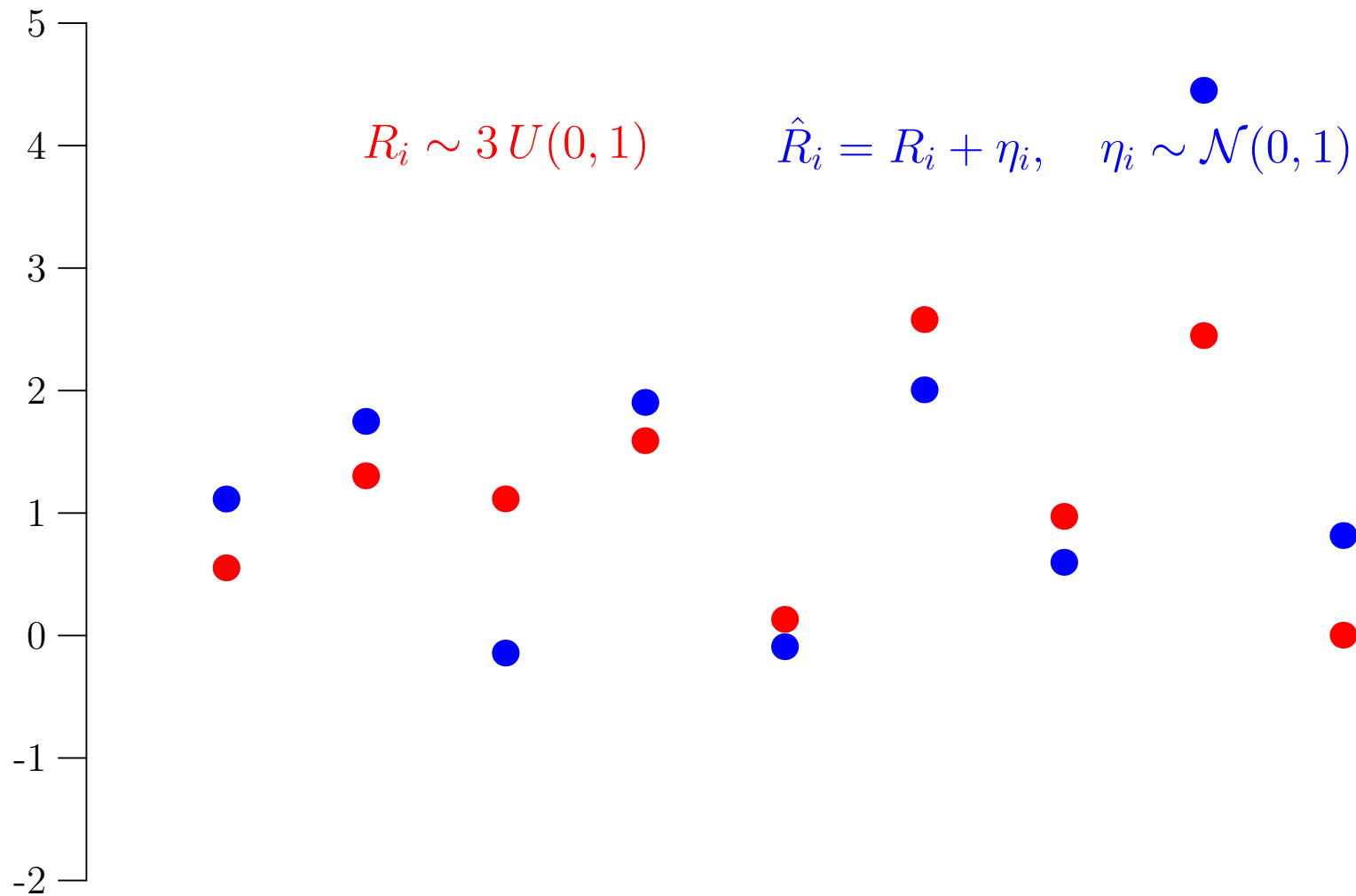
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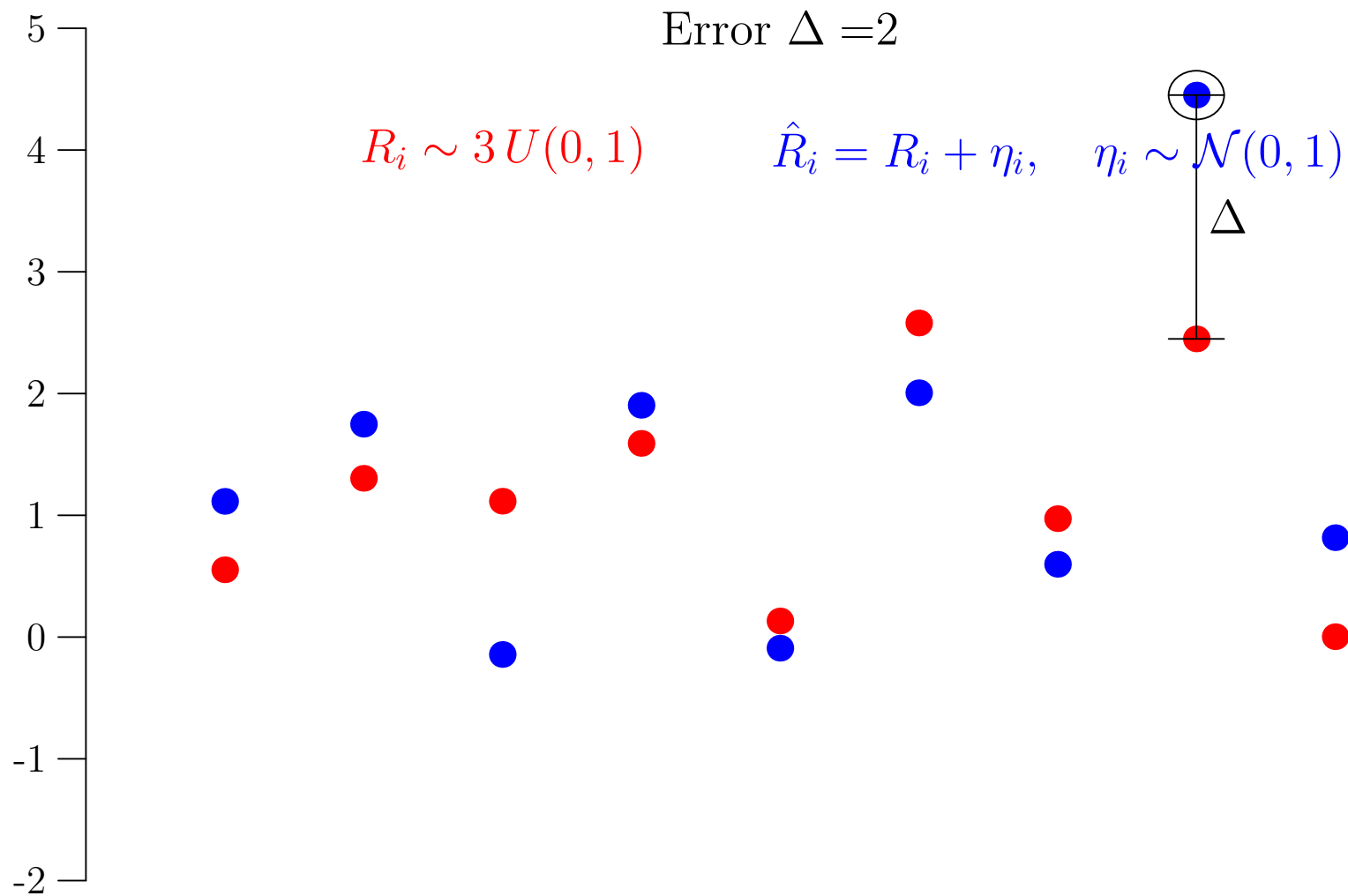
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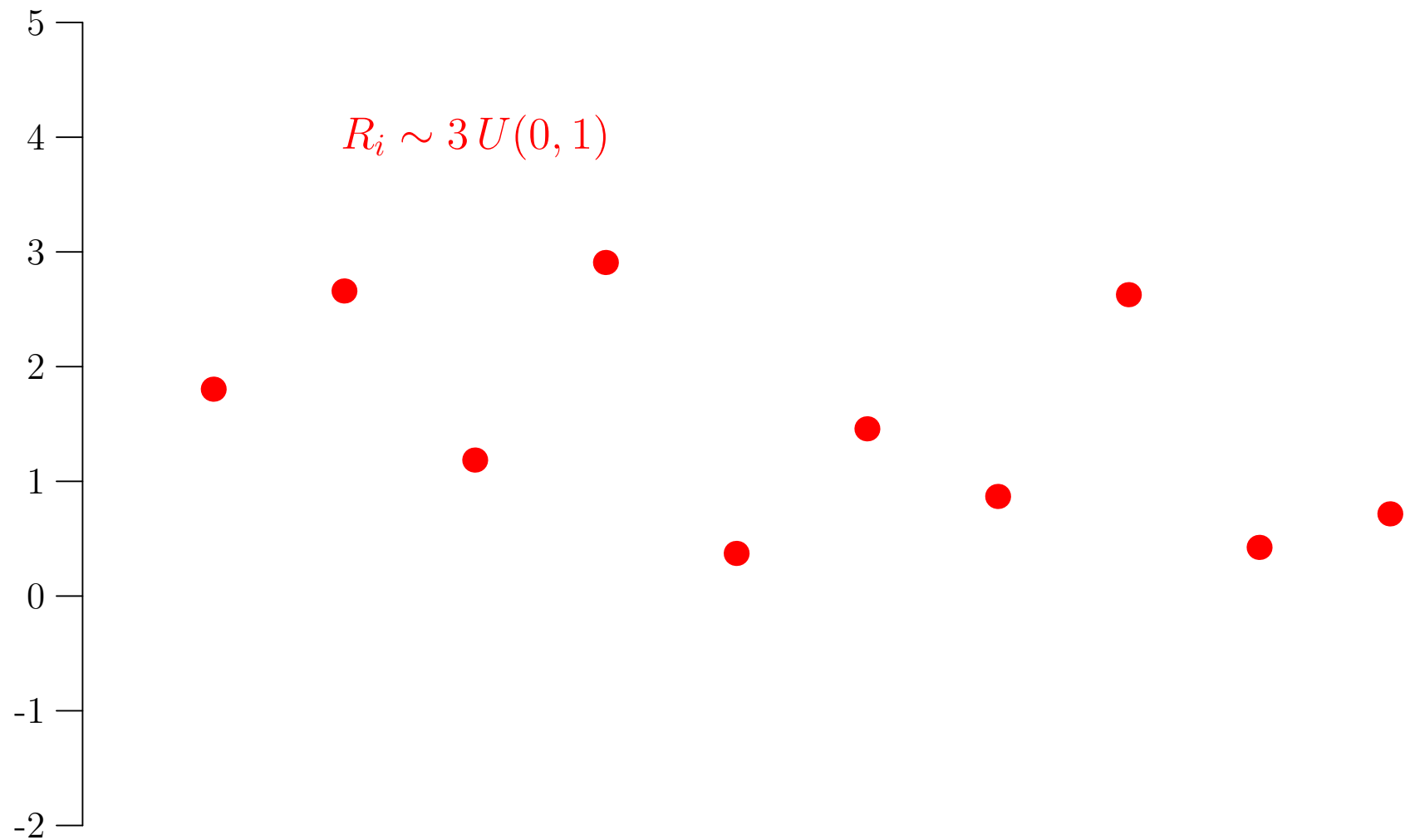
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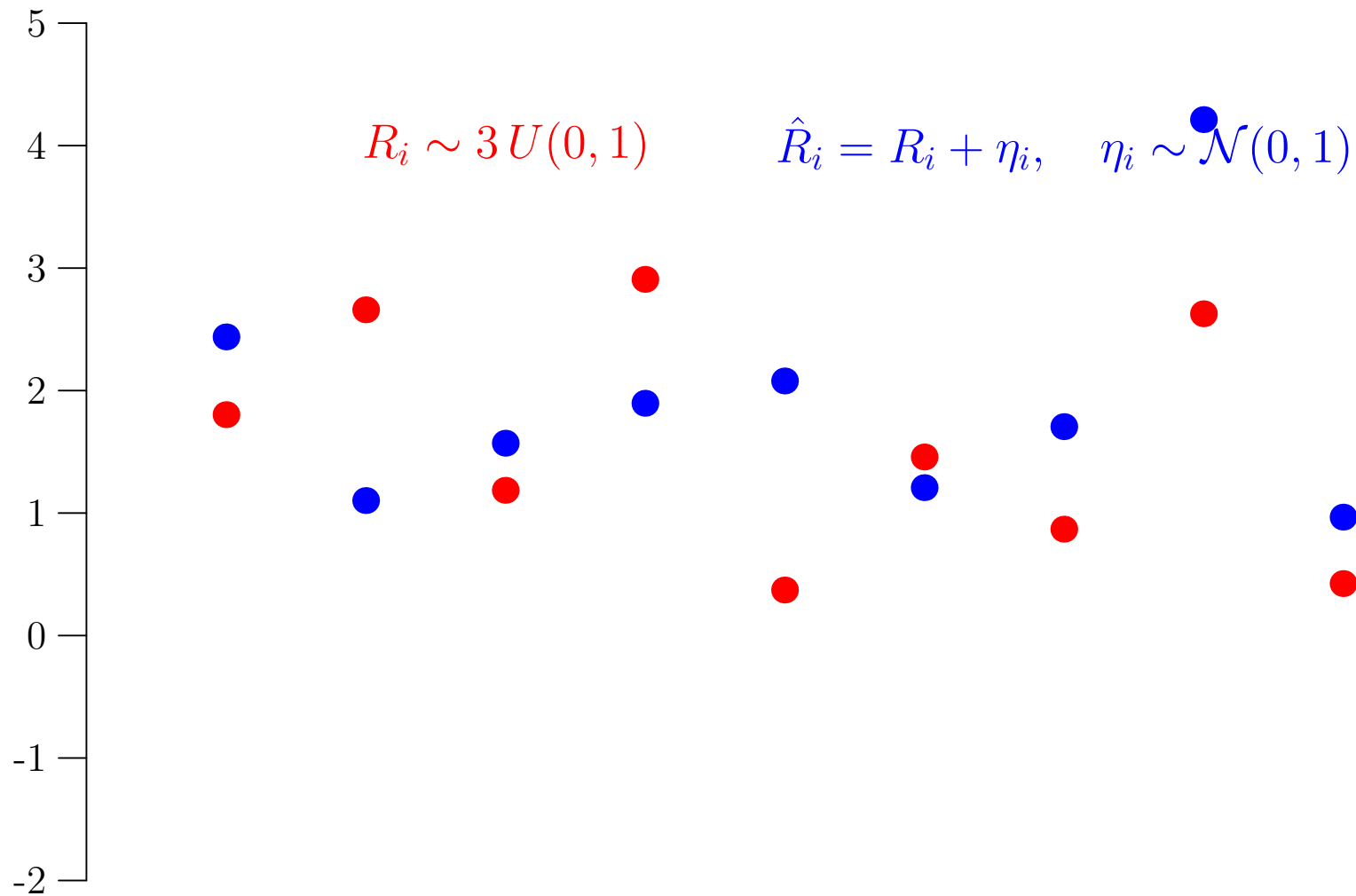
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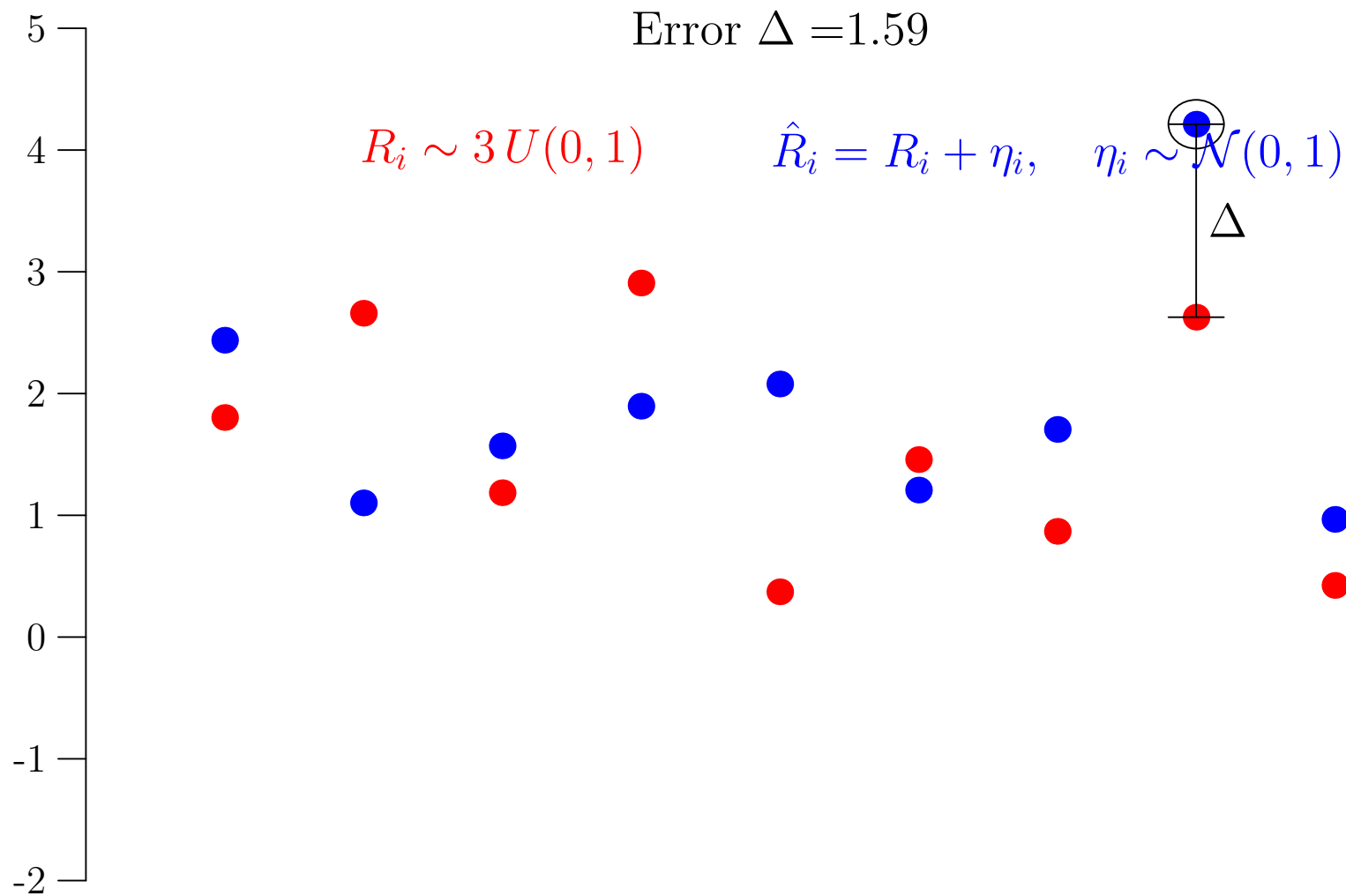
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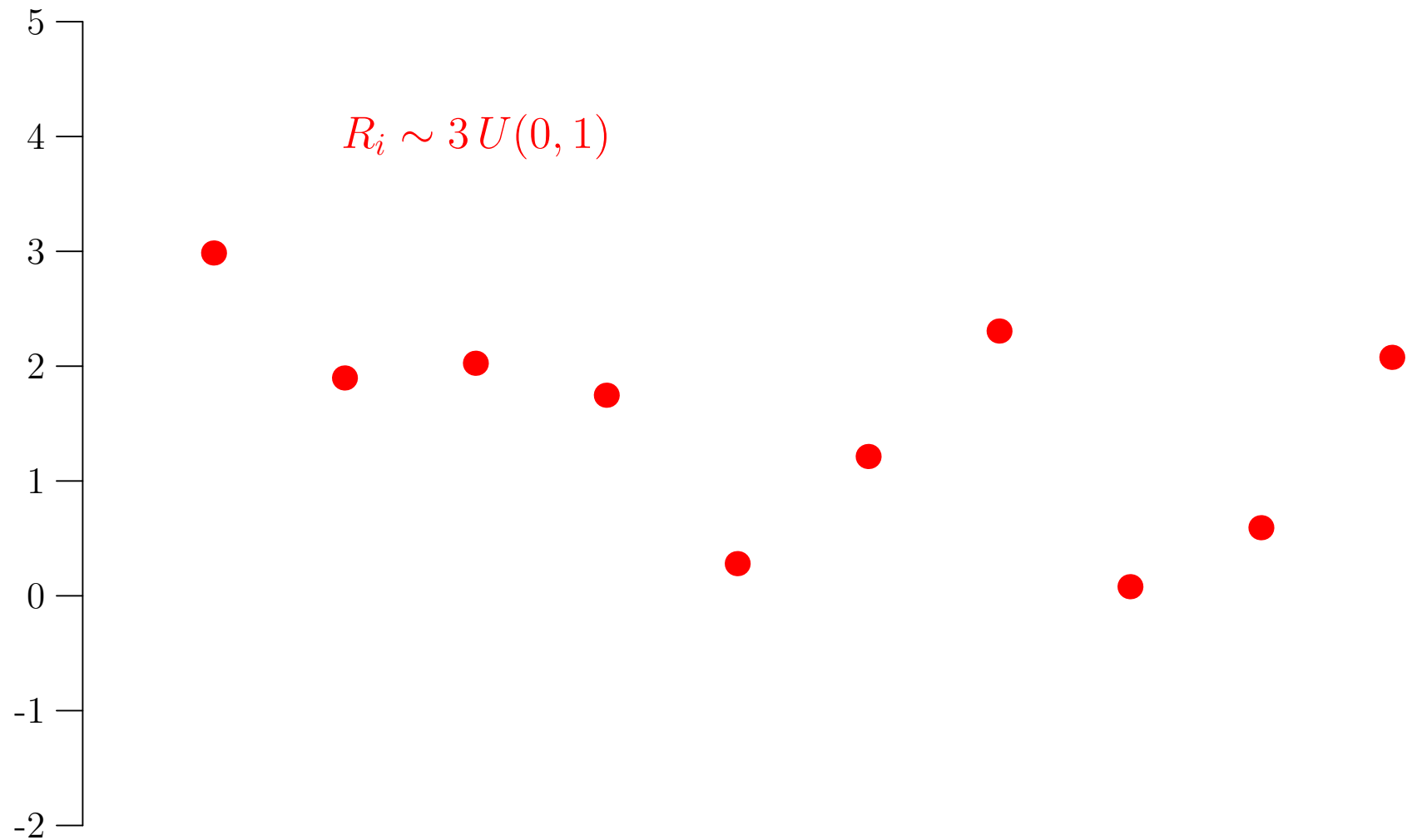
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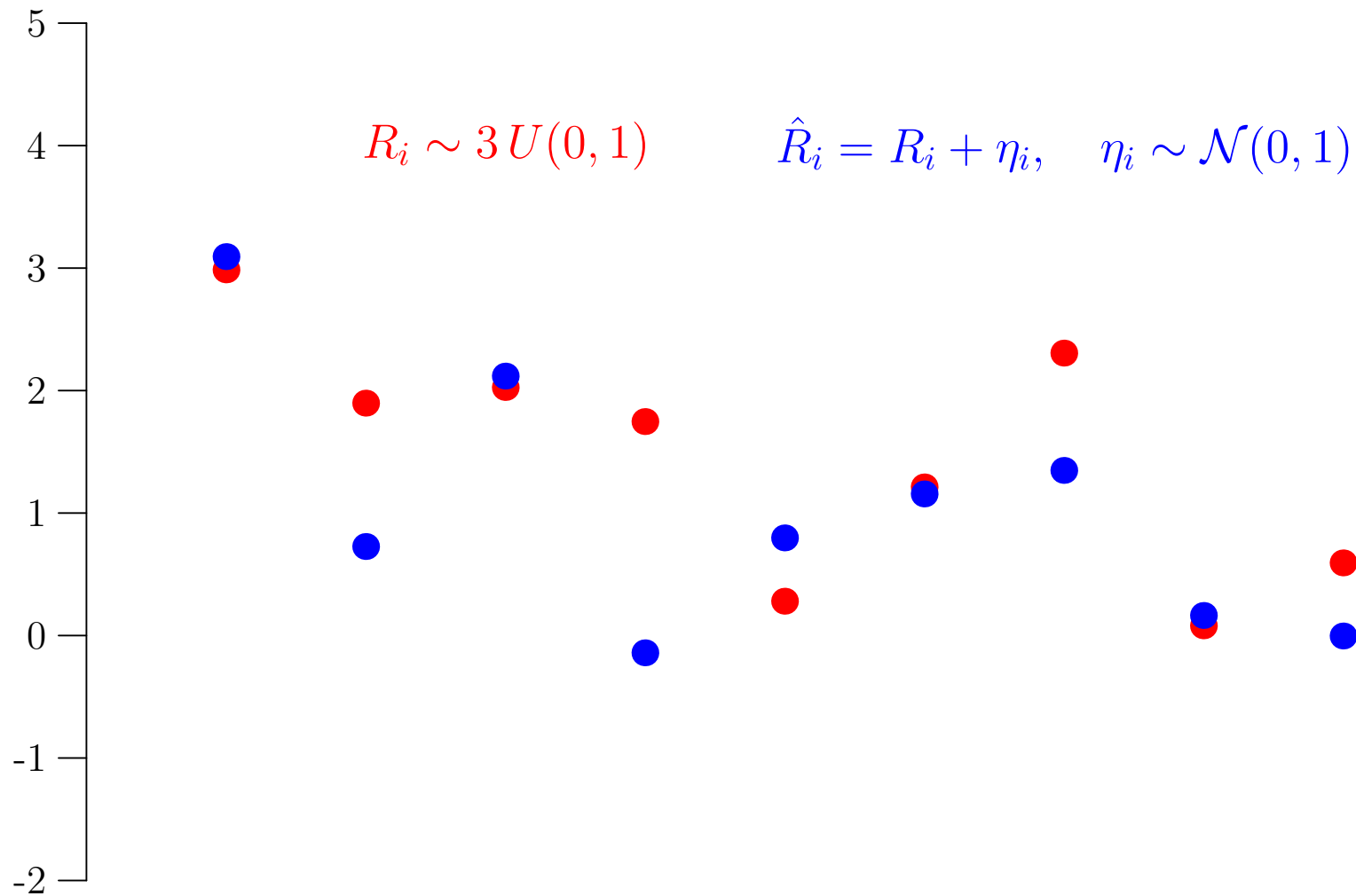
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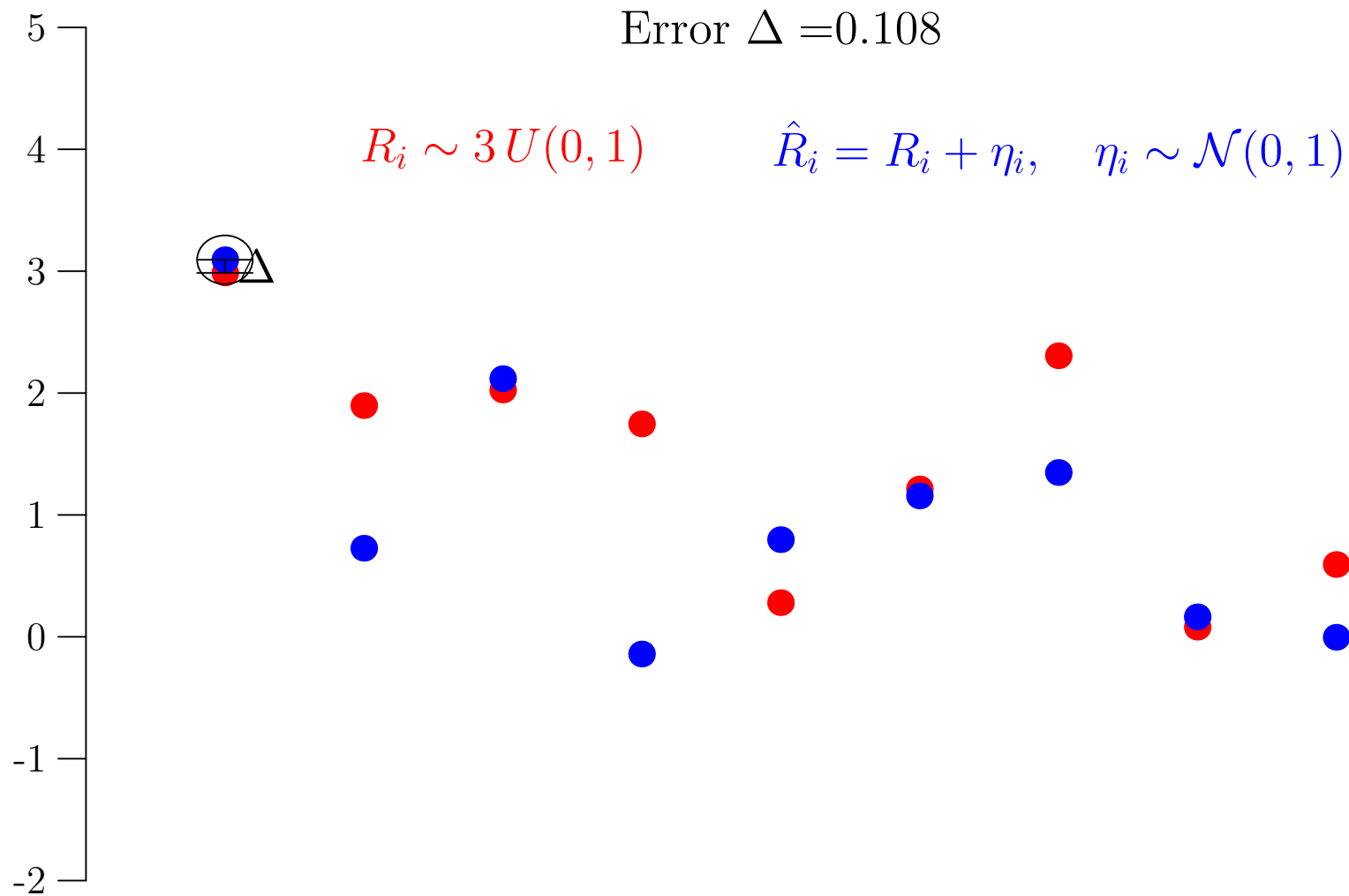
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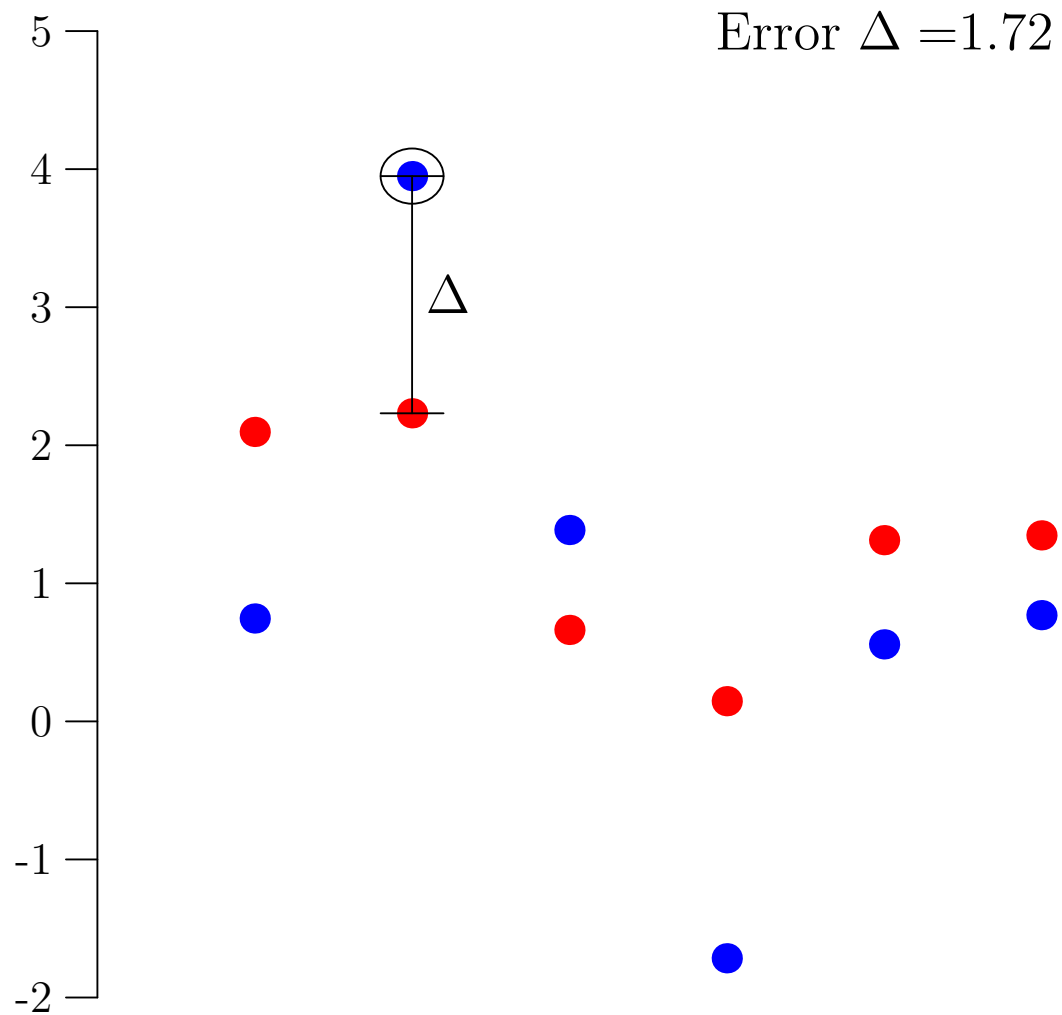
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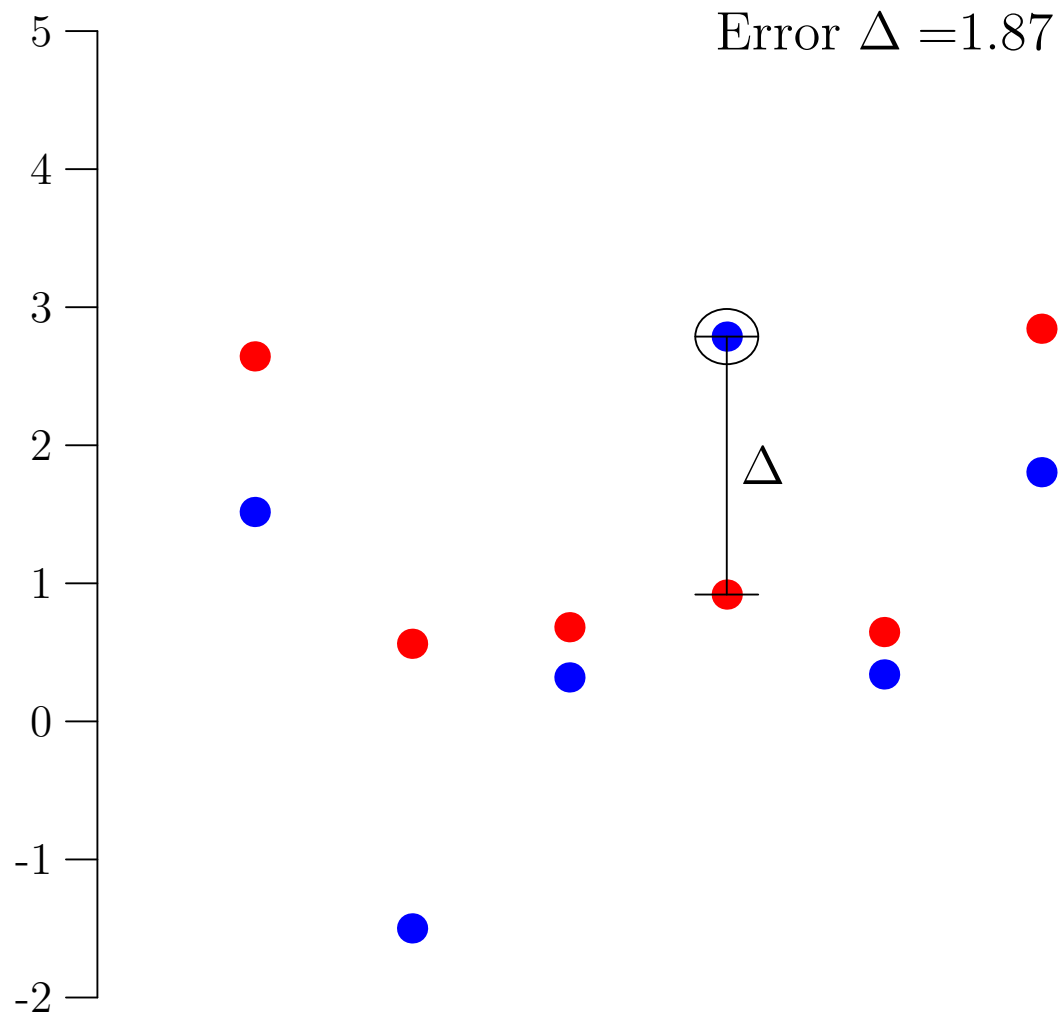
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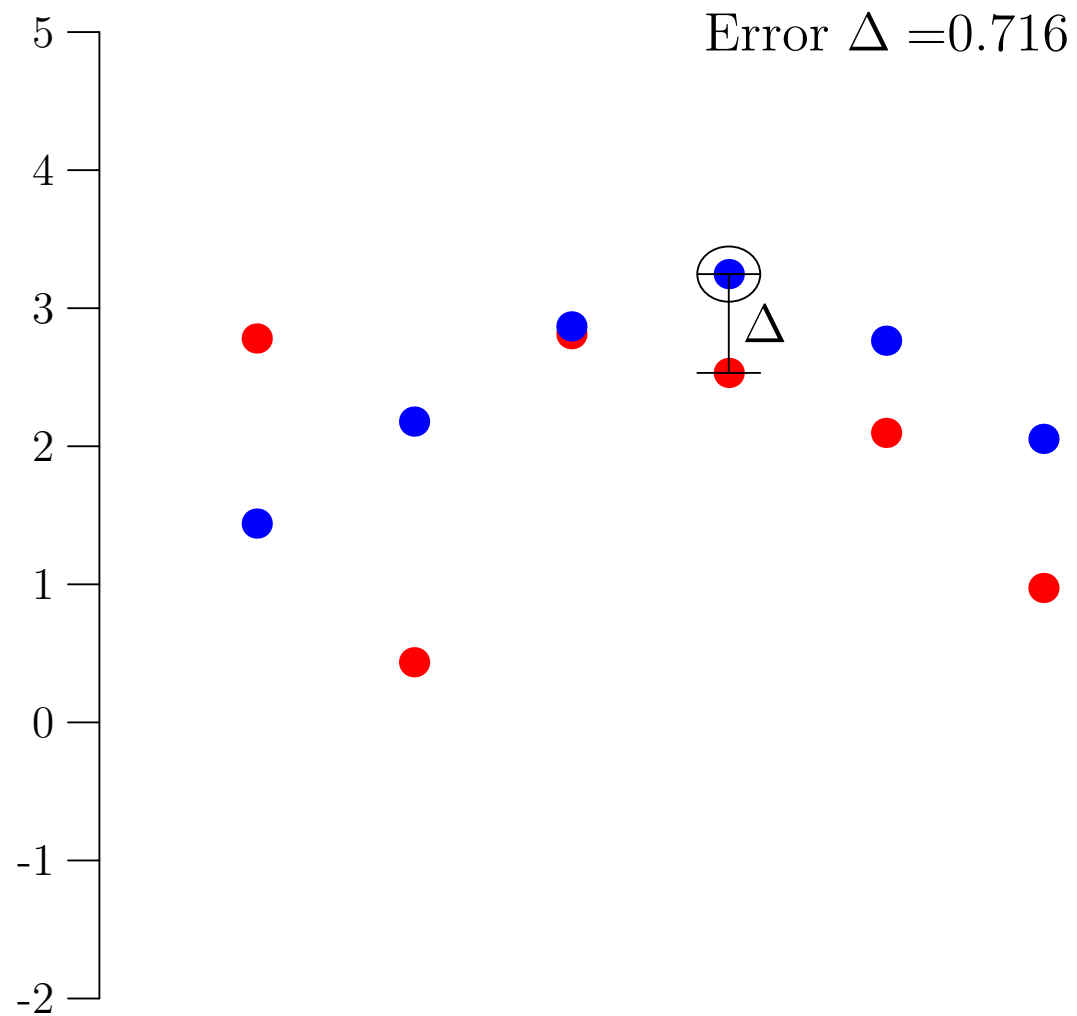
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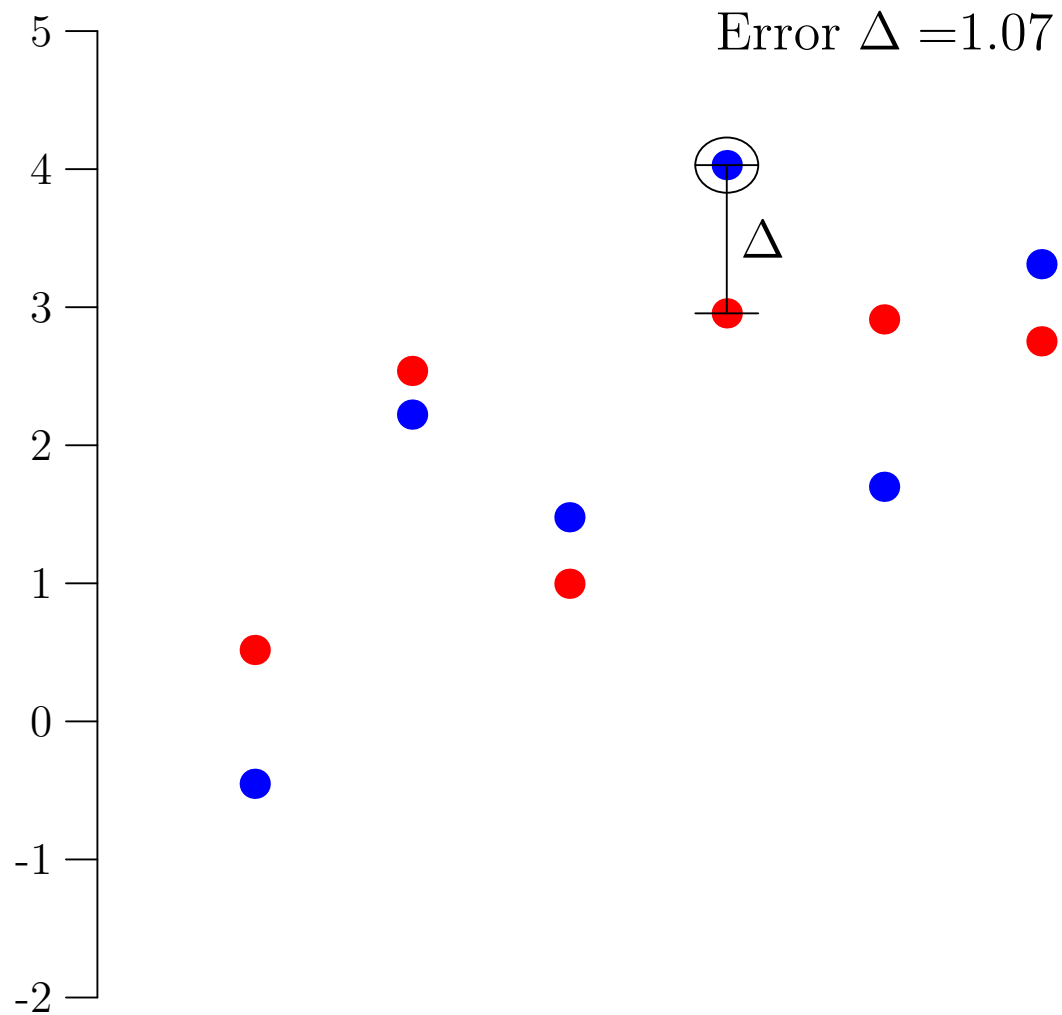
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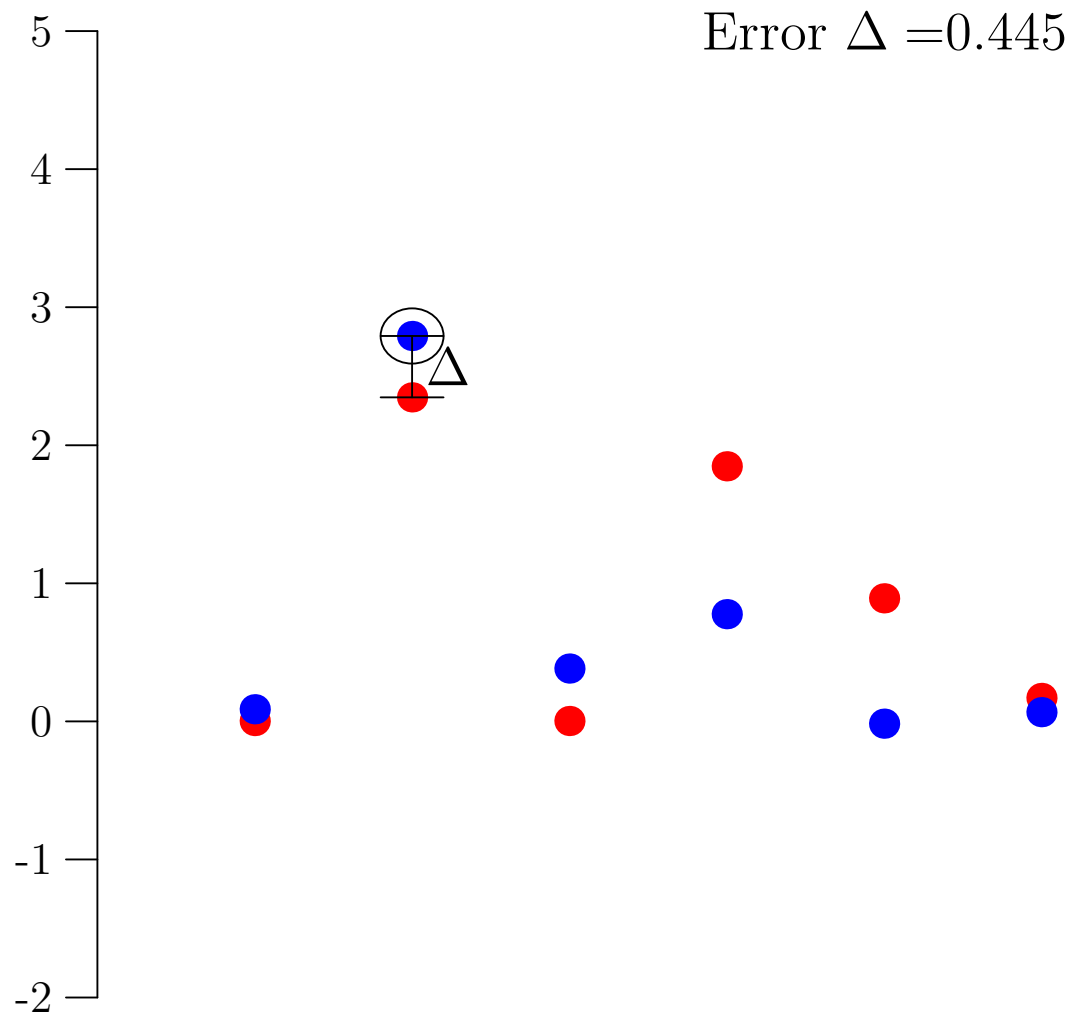
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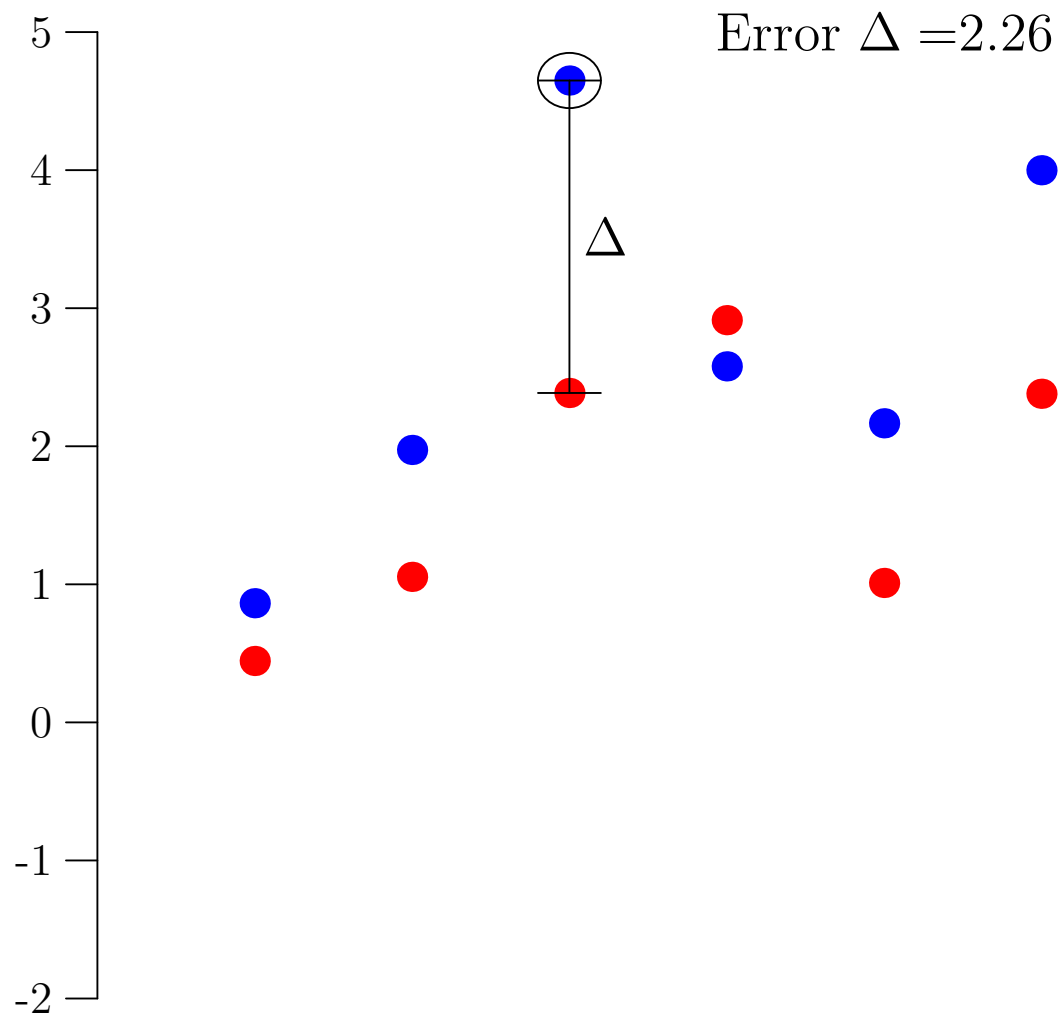
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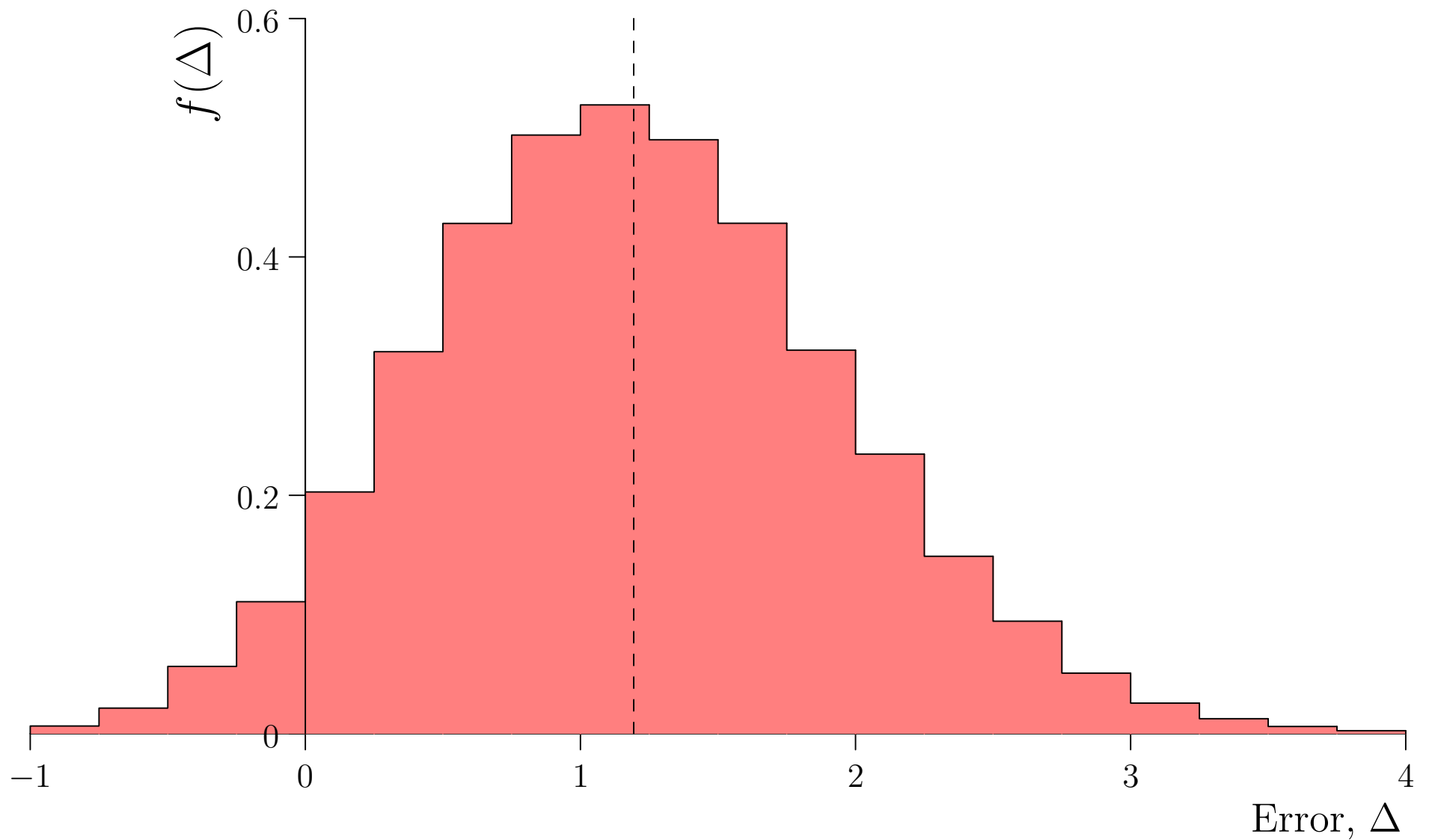
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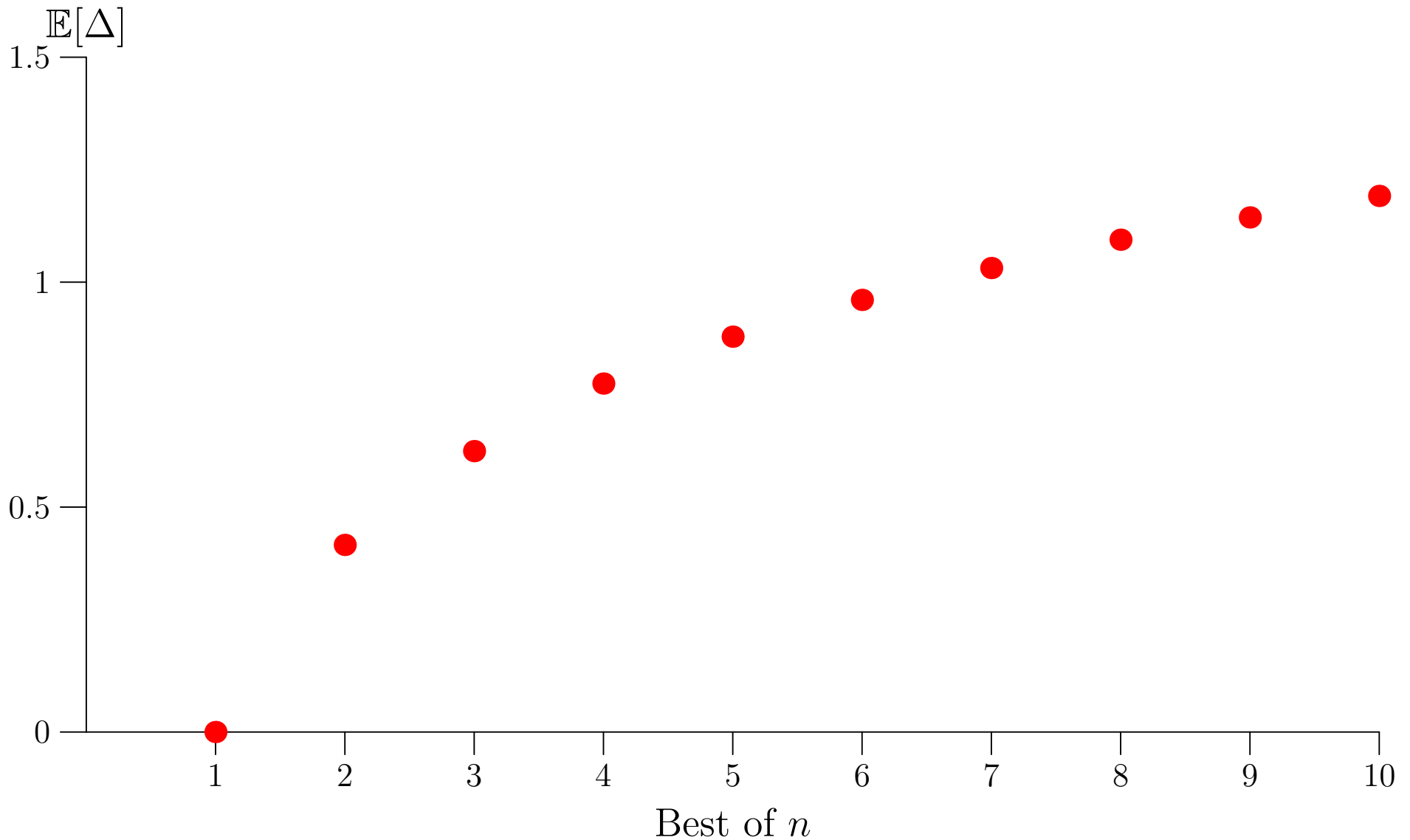
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Overfitting the Test Set

- If you try to improve your performance by regularly testing on your test set you quickly overfit to the test set
- In many Kaggle competitions there is a test set provided early on which is used to maintain a leader board
- At the end of the competition, the competitors are tested on a new test set and the order of the leader board is radically changed
- You can be honest by using a separate **validation set**

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- **Lesson 1: Beware of over-fitting on the test set**

CIFAR-10

Downloading data from <https://www.cs.toronto.edu/~kriz/cifar-10-python.tar.gz>
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- 10 class problem with colour images of size 32×32
- MLPs struggle

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Dodgy Test Sets

- 10 or so years after CIFAR-10 was produced Benjamin Recht *et al.* constructed a new test set for CIFAR-10 using the same protocol
- The performance of the best performing models dropped from 4% error to 10% error
- A lot of this was down to over-fitting the test set
- But it also turned out there was some contamination in the original test-set with images in the training set appearing in the test set
- There may have been some *shortcut learning* (perhaps trucks were taken predominantly from a web site with images from mobile phones, birds being taken with long lens)

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- **Lesson 2: Be suspicious of data sets**

ImageNet



- ImageNet for several years was considered the Gold Standard for measuring image classification performance
- The ImageNet competition started in 2009 was instrumental in starting the deep learning revolution with AlexNet beating the competition in 2011 by 10% and ushered in GPUs
- It used large images (224×224 or 256×256) with 1000 classes and 1.3 million training examples

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- The dataset requires a lot of compute to train a network: practitioners often use mini-ImageNet versions to reduce heating up the planet
- It has oddity, humans exist in the image, but are ignored (there is no human category)
- There is no perfect training set, but ImageNet pushed the field more than most

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Data Set Contamination

- Because training large models is time consuming it is regular practice to build on fine-tuned large (foundation) models
- This is especially true in LLMs
- Sometimes it is possible to get **surprisingly** good performance on tasks because the backbone model was trained on that task
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- **Lesson 3: Avoid data set contamination**

Short Cut Learning

- The field is full of datasets where there is a short-cut for making a classification
- Classic examples include X-ray diagnostic systems that pick-up on post-operative artifacts showing the patient had a disease
- Machine learning will pick-up on any consistent signal that separates data classes, even when these are not what you want to learn

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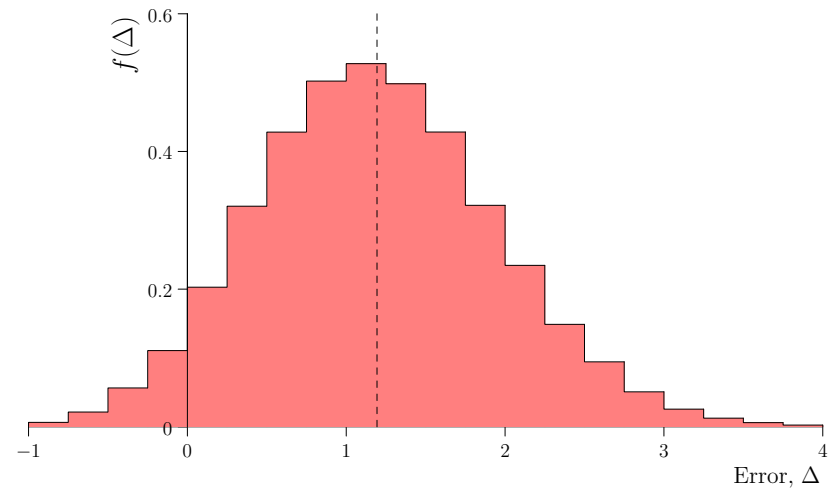
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- **Lesson 4: When results seem too good to be true, often they are too good to be true**

Outline

1. Data Sets
2. **Reproducibility**
3. Metrics



Look at My New Model

- Machine Learning has always be plagued by people developing new models
- Sometimes these model turn out to change everything (AlexNet, Transformers)
- More often these models are claimed to beat an existing model, but sink without leaving a bubble
- There are many causes

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One-trick Pony

- Models are often tested on a single problem
- Either they are compared against a weak baseline
- Or they have been optimised, while the baseline used out-of-the-box hyper-parameters
- Referees have become very wary of these claims and will often insist on testing on multiple datasets, against strong baselines that are equally optimised
- This is boring to do, but comes from experience

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- **Lesson 5: Be rigorous in comparing with others**

Irreproducibility

- Many results claimed in papers cannot be reproduced
- A classic example happened to a much vaunted paper by Geoff Hinton about an architecture known as Capsules claiming a SOTA performance of 0.25% test error on MNIST
- This was never reproduced
- ML went through an existential crisis about reproducing results
- It is now standard practice to publish code to allow others to quickly verify your results

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- **Lesson 6: Make your work reproducible: prepare to make your code public**

Claims on How it Works

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- **Lesson 7: Perform ablation studies where meaningful**

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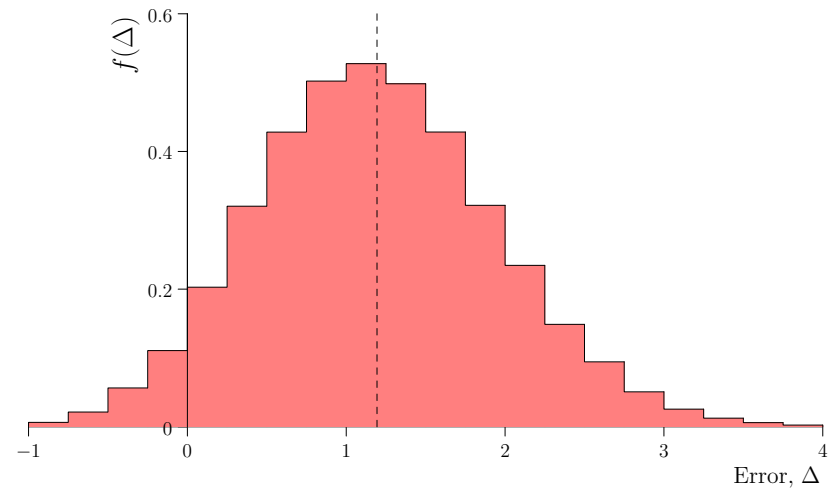
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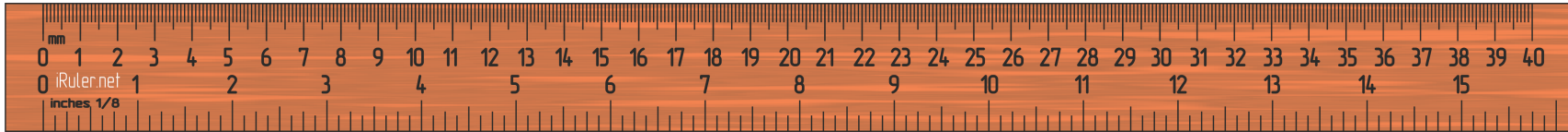
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Outline

1. Data Sets
2. Reproducibility
3. **Metrics**

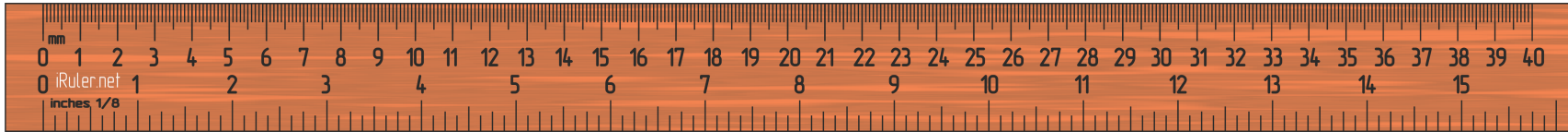


Metrics



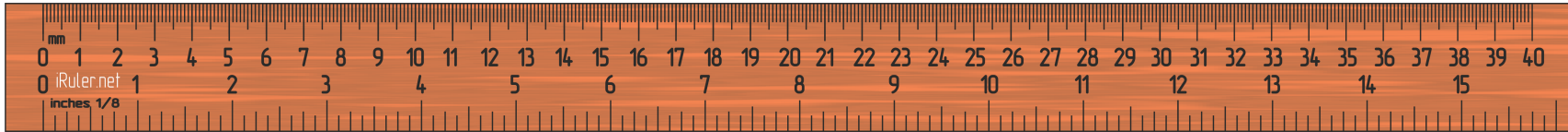
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Choosing Good Metrics

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- **Lesson 10: Help readers to understand your metrics (should it be maximised or minimised? What values are acceptable/good/bad?)**

Human Metrics

- For some problems a suitable metric is human evaluation
- When this does not involve specialists then there are many crowd-sourcing companies that can do this
- This is expensive, so practitioners often fall-back of automatic metrics
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Don't Necessarily Follow the Trend

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- **Lesson 11: When everyone is concentrated on improving one metric, think is there another metric that measures something that may be more important**

Presenting Metrics

- Measuring performance comes with errors
- There may be systematic error due to the biases in your test set
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- **Lesson 12: Always compute estimated errors in the mean**

Errorbars for Bernoulli Trials

- If you have some binary measurement and you want to estimate the probability of success, p , then your estimated error in mean is

$$\Delta = \sqrt{\frac{p(1-p)}{n}}$$

- Of course, you don't know p , but the worst it gets is when $p = \frac{1}{2}$ (giving errors in the mean of $\Delta = \frac{1}{2\sqrt{n}}$)
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- You can get a tighter bound by using $p \approx \hat{p} = s/n$
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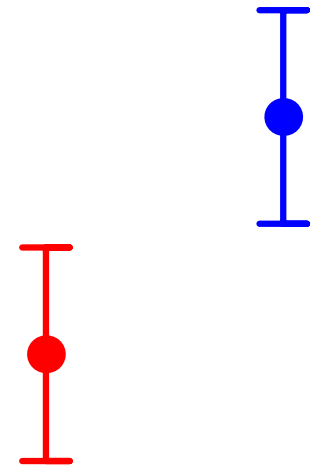
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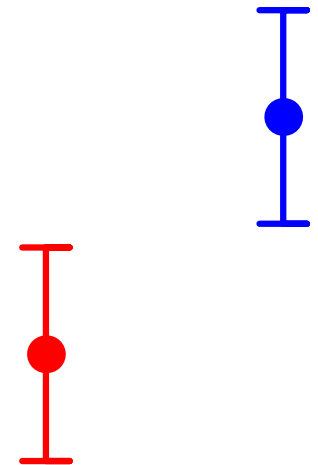
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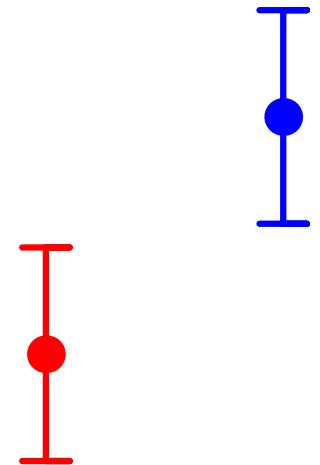
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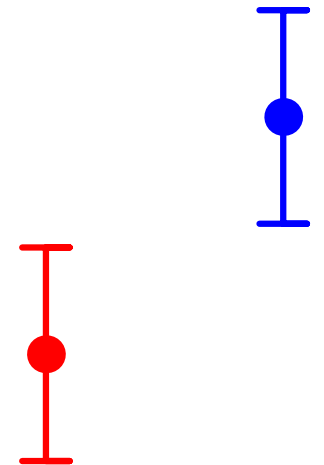
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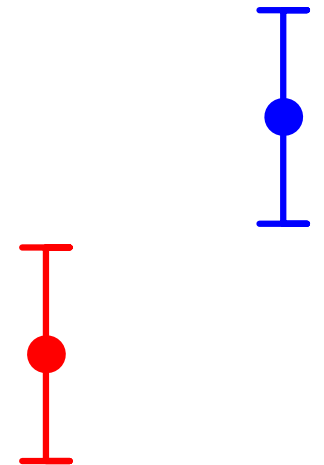
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- **Lesson 13: Compute statistical significance when necessary**
- **Lesson 14: Statistical significance doesn't necessarily mean this is a difference you care about**



Standard Deviation and Errors In Mean

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- You can compute a p -value by simulating the chance event and seeing what proportion of the time you exceed your result
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- **Lesson 15: Think carefully about how to present your results convincingly**

Summary

- There are a lot of lessons to learn from past malpractice
- Following some of them (testing of different datasets, multiple models, providing code, optimising other people's code) is a pain to do
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- Following best practice will ultimately make you a better scientist

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- **Lesson 16: Using robust commonsense can go a long way to avoid common errors**